

All-in-One. No External Antenna.



ODU302 4G Industrial Outdoor Router

· 4G LTE

· All-in-One

· Outdoor IP65

· Link Backup

1. Product Overview

The ODU302 is a fully integrated 4G industrial outdoor router that combines a high-performance router and high-gain antennas in one IP65 enclosure. Designed for ATMs, EV chargers, vending machines, and outdoor control cabinets, it mounts directly on top of equipment — eliminating the need for indoor routers, long coax feeders, and separate outdoor antennas. The ODU302 simplifies installation to a single Ethernet cable with PoE support, delivering enterprise-grade connectivity with minimal deployment complexity.

Key Features

Value Dimension	Description
Router + Antenna Integrated	One SKU, one top-mount installation — no separate outdoor antenna to source or match
Zero Feeder Loss	RF module adjacent to the antenna eliminates coax attenuation for stronger 4G signal
PoE Single-Cable	IEEE 802.3af PoE input delivers power and data over one Ethernet cable; DC 9–36 V also available
Dual SIM 4G LTE Redundancy	Dual Nano SIM failover with intelligent switching for 24/7 uptime
Enterprise Security	SPI firewall, ACL, IPsec, OpenVPN, WireGuard, ZeroTier
Cloud-Managed	InHand Device Manager for remote monitoring, configuration, and OTA firmware upgrades
Built for Outdoors	IP65 protection, -20 °C ~ +70 °C operating range

Typical Application Scenarios

Financial Self-Service (ATM)

Limited cabinet space and long feeder runs degrade 4G signal. The ODU302 top-mount design frees the cabinet interior and delivers zero-feeder-loss RF performance for more stable transactions.

Smart Energy (EV Charging)

Outdoor exposure, waterproof wiring, and harsh climate are constant challenges. IP65 + wide-temp rating, PoE single-cable design, and Device Manager fleet ops keep charging networks online.

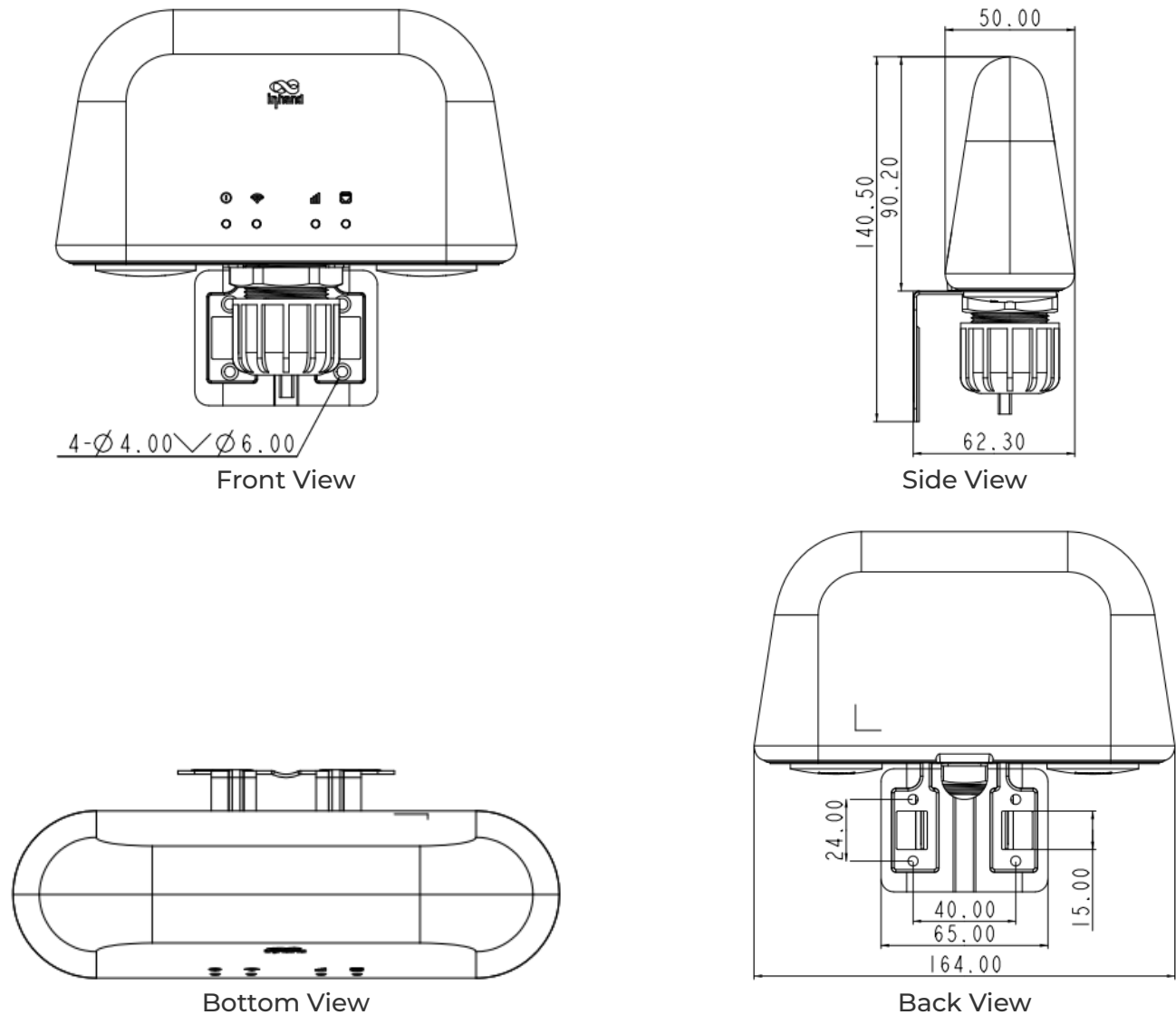
Self-Service Retail (Vending)

Fast rollout, vandal-resistant placement, and minimal internal cable clutter. One integrated unit means no separate outdoor antenna sourcing and reduced in-cabinet wiring.

Industrial Control (Outdoor Cabinets)

Multiple penetrations for power, Ethernet, and coax create sealing complexity. The ODU302 needs only one Ethernet drop into the cabinet, plus 2× configurable digital I/O for signal acquisition and control.

Product Dimensions



Notes:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. Dimensions (L × W × H): 164 × 50 × 129.7 mm (6.46 × 1.97 × 5.11 in).
3. All dimensions are approximate and **for reference only**.
4. Drawings **must not be used for manufacturing**.
5. Specifications may change without prior notice.

2. Product Innovations

Innovation 1: Router + Antenna in One Top-Mount Unit

The router is built directly into the antenna enclosure and installs on top of ATMs, EV chargers, and control cabinets. No separate outdoor antenna to source or match. No wall penetrations for feeder cable. One SKU, one mount — dramatically reducing on-site labor and system integration complexity.

Innovation 2: Zero Feeder Loss, Stronger Signal

With the RF module and antenna integrated as one, the signal travels with zero feeder attenuation — a major advantage on high LTE bands where coax loss is significant. You also eliminate the cost and sourcing effort of a separate outdoor-rated antenna, since the ODU302 ships with optimized built-in antennas and IP65 protection.

Innovation 3: PoE Single-Cable Deployment

The ODU302 accepts IEEE 802.3af PoE input. From the top-mounted unit to the cabinet interior, only one Ethernet cable is needed for both power and data. The wiring bundle shrinks from "power + Ethernet + feeder" down to a single drop. DC 9–36 V terminal input is also available as an alternative.

Innovation 4: Built for Demanding Outdoor Environments

The IP65 enclosure is rated against dust ingress and high-pressure water jets, with an operating temperature range of -20 °C to +70 °C. Tested to IEC 60068-2-27 shock, IEC 60068-2-6 vibration, and EN 61000-4-x EMC Level 2. The unit installs directly on top of the equipment with full outdoor exposure — no extra hood required for standard rain and dust.

Innovation 5: Proven IR302 Software Platform

ODU302 runs the same software stack as the field-proven InRouter302. Dual-SIM failover, enterprise firewall, VPN, link backup, digital I/O, and InHand Device Manager cloud management — all carried over with zero learning curve. Existing IR302 customers can migrate their configurations and workflows seamlessly.

3. Hardware Specifications

3.1 Hardware Overview

Parameter	Specification
Processor	580 MHz
Memory	128 MB DDR2 SDRAM
Storage	32 MB SPI NOR Flash
4G Module	4G LTE Cat.4, DL 150 Mbps / UL 50 Mbps
Recommended Users	32
Ethernet Ports	2 × 10/100 Mbps RJ45, WAN/LAN switchable. LAN2 supports PoE
SIM Slots	2 × Nano SIM (drawer-style), dual SIM failover
Digital I/O	2 × configurable DI or DO
4G Antennas	2 × built-in 4G antennas (ANT_4G_1, ANT_4G_2)
Wi-Fi Antennas	1 × built-in Wi-Fi antenna
Wi-Fi	IEEE 802.11 b/g/n (Wi-Fi 4), 2.4 GHz, up to 150 Mbps

Parameter	Specification
Wi-Fi TX Power	802.11b: 16 dBm $\pm$ 2 dBm (11 Mbps) 802.11g: 16 dBm $\pm$ 2 dBm (54 Mbps) 802.11n@2.4 GHz HT20 MCS7: 16 dBm $\pm$ 2 dBm 802.11n@2.4 GHz HT40 MCS7: 16 dBm $\pm$ 2 dBm
LED Indicators	System, Wi-Fi, Cellular, Ethernet
Reset Button	Pinhole reset button
Power Supply	DC 9–36 V (4-pin 2×2 combined power & I/O terminal, over-current / reverse-polarity protected) IEEE 802.3af PoE input (Mode A or Mode B)
Standby Power	90~110 mA @ 12 V
Operating Power	280~300 mA @ 12 V
Peak Power	340~360 mA @ 12 V
Protection Rating	IP65
Enclosure Material	Industrial-grade plastic
Operating Temperature	-20 °C ~ +70 °C (-4 °F ~ +158 °F)
Storage Temperature	-40 °C ~ +85 °C (-40 °F ~ +185 °F)
Relative Humidity	5% ~ 95%, non-condensing
Dimensions	164 × 50 × 129.7 mm (6.46 × 1.97 × 5.11 in)
Weight	Approx. 500 g (1.10 lb)
Mounting	Wall, pole, top-mount

3.2 Interface Overview

RJ45 Ports

Port	Default	Description
WAN/LA N1 (Left)	WAN/L AN	10/100 Mbps, software switchable
LAN2 (Middle)	LAN	10/100 Mbps, PoE PD (Powered Device), accepts power on data pairs (1/2, 3/6) for Mode A, or on spare pairs (4/5, 7/8) for Mode B. IEEE 802.3af compliant

SIM Card Slots

Slot	Description
SIM1 (Top)	Primary SIM card (Nano SIM, drawer-style)
SIM2 (Bottom)	Backup SIM card (Nano SIM, drawer-style)

Reset Button

- Hardware RST: Pinhole reset button

Power & I/O Terminal (4-Pin, 2×2)

Position	Top Pin	Bottom Pin
Left	IO1 (DI/DO1, configurable)	GND
Right	IO2 (DI/DO2, configurable)	DC 9–36 V Input (Positive)

3.3 Environment & Reliability

Test Item	Standard	Level/Result
ESD	EN 61000-4-2	Level 2
Radiated RF	EN 61000-4-3	Level 2
EFT/Burst	EN 61000-4-4	Level 2
Surge	EN 61000-4-5	Level 2
Conducted RF	EN 61000-4-6	Level 2
Oscillatory Wave	EN 61000-4-12	Level 2
Power Frequency Magnetic	EN 61000-4-8	400 A/m ($\geq$ Level 2)
Shock	IEC 60068-2-27	Compliant
Vibration	IEC 60068-2-6	Compliant
Free Fall	IEC 60068-2-32	Compliant

4. Network Connectivity

4.1 Cellular Network

- **Network Standards:** GSM/GPRS/EDGE, UMTS/HSPA+/EVDO, TD-SCDMA, FDD LTE/TDD LTE
- **Network Access:** APN, VPDN
- **Authentication:** CHAP/PAP
- **Connection Modes:** Always-on, on-demand, manual dial
- **Activation:** Data trigger, SMS trigger

4.2 SIM Management

- **Dual-SIM Support:** Drawer-style dual Nano SIM card slots
- **Dual SIM Failover:** Primary/backup SIM auto-switch support
- **Smart Switching:**
 - Automatic SIM switch on dial failure
 - Automatic switch when signal falls below threshold
 - Configurable primary/backup card and max dial attempts
- **SIM Binding:** ICCID binding, PIN code management

4.3 Wired Network

- **WAN Protocol:** Static IP, DHCP, PPPoE
- **LAN Protocol:** ARP, Ethernet
- **Interface Mode:** 2 × RJ45 ports, WAN/LAN configurable
- **NAT:** Network address translation (NAPT)

4.4 Wi-Fi 4 Wireless Access

- **Standard:** IEEE 802.11 b/g/n (Wi-Fi 4)
- **Frequency:** 2.4 GHz single-band
- **Max Data Rate:** 150 Mbps
- **Modes:** Access Point (AP), Client (STA)
- **Security:** Open, Shared Key, WPA/WPA2
- **Encryption:** WEP/TKIP/AES

5. Link Management & Backup

5.1 Intelligent Link Backup

- **Backup Modes:** Cellular dial-up and WAN port can serve as primary/backup for each other
- **Probe Methods:** ICMP/TCP/DNS link quality probing (configurable)

- **Switching Policies:** Automatic failover on link failure, auto-redial on disconnection

5.2 VRRP Hot Backup

- **Protocol Support:** VRRP (Virtual Router Redundancy Protocol)
- **Use Case:** Dual-device hot standby for improved business continuity

5.3 Link Quality Assurance

- **Link Detection:** Heartbeat packet monitoring, auto-reconnect on disconnection
- **Dual SIM Switching:** Automatic backup SIM switch on cellular link or dial-up anomaly
- **Built-in Watchdog:** Device self-check with fault auto-recovery

6. Security Features

6.1 Firewall

- **Stateful Inspection:** Stateful packet inspection (SPI)
- **DoS Protection:** Denial-of-service attack prevention
- **Access Control:** Access Control List (ACL)
- **Content Filtering:** URL-based content filtering
- **Port Mapping:** Port forwarding, virtual IP mapping
- **IP-MAC Binding:** IP-MAC binding; multicast/Ping packet filtering
- **NAT:** NAT support

6.2 VPN Support

- IPsec VPN (IKEv1, IKEv2)
- L2TP VPN (client)
- PPTP VPN (client)
- GRE Tunnel
- OpenVPN (client/server)

- WireGuard
- ZeroTier
- CA digital certificate support

6.3 Traffic Shaping

- **Bandwidth Control:** Interface-level bandwidth limiting
- **IP Rate Limiting:** Per-IP upload/download rate limiting
- **QoS Policy:** Bandwidth and rate limit policies

7. Local Network Services

7.1 DHCP Service

- **DHCP Server:** Dynamic IP address assignment
- **DHCP Relay:** Cross-subnet DHCP relay
- **DHCP Client:** Client address acquisition

7.2 DNS Service

- **DNS Relay/Proxy:** Local caching and forwarding
- **Custom DNS:** Configurable DNS server

7.3 Dynamic DNS (DDNS)

- **Multi-Provider Support:** Major DDNS service providers
- **Automatic Updates:** Automatic domain resolution update on WAN/cellular IP change

7.4 Static Routing

- **Route Entries:** Static route configuration
- **Default Routes:** Cellular/WAN default route support

7.5 IP Passthrough

- **Bridge Mode:** WAN/dial-up IP pass-through to downstream device
- **Flexible Configuration:** Passthrough mode on dial-up and WAN ports

8. Cloud Management & Operations

The ODU302 is powered by the InHand Device Manager cloud management platform, enabling unified remote operations for large-scale distributed outdoor devices.

8.1 Device Manager Cloud Management

Feature	Description
Device Onboarding	Device Manager cloud platform and custom server support
Remote Management	Device status query, remote configuration push, firmware upgrade, remote reboot
Batch Operations	Mass device configuration and maintenance
SNMP	SNMP v1/v2c/v3 with Trap support
Traffic Management	Traffic thresholds, statistics, and alarms
Alert Types	System reboot, LAN port up/down, traffic alarm, SIM card fault
Notification	Email alerts (configurable)

9. System Management & Maintenance

9.1 Device Management

- **Web Management:** Graphical web configuration interface
- **CLI Management:** Telnet, SSH, Console
- **Configuration Import/Export:** Backup and restore
- **Factory Reset:** Pinhole reset or web factory reset
- **Firmware Upgrade:** Web or Device Manager remote upgrade

9.2 Logs & Diagnostics

- **System Logs:** Local syslog, remote log, serial log export; persistent critical logs on power loss
- **Diagnostic Tools:** Ping (network connectivity), Traceroute (route tracing), Network speed test (link performance)

9.3 SMS Functions

- **Status Query:** SMS query device status
- **Remote Reboot:** SMS-triggered device reboot

9.4 Other System Capabilities

- **Dial-on-Demand:** On-demand dial; data/SMS activation
- **Status Query:** System status, modem status, network connection status, routing status
- **Scheduled Maintenance:** NTP time synchronization; auto-recovery on fault (built-in watchdog)

10. Ordering Information

10.1 Model Code Structure

Model code: ODU302-

Field	Code	Description
Type & Module	CNC4	China, 4G LTE Cat.4, Wi-Fi, I/O
	NAC4	North America, 4G LTE Cat.4, Wi-Fi, I/O
	EUC4	Europe/APAC, 4G LTE Cat.4, Wi-Fi, I/O

10.2 Product Models

Model	Region	Description
ODU302-CNC4	China	LTE-FDD: B1/B3/B5/B8 LTE-TDD: B34/B38/B39/B40/B41 WCDMA: B1/B8 GSM: B3/B8 CDMA: BC0 TD-SCDMA: B34/B39 Wi-Fi: 802.11 b/g/n I/O: 2× configurable DI/DO
ODU302-NAC4	North America	LTE-FDD: B2/B4/B5/B12/B13/B14/B66/B71 WCDMA: B2/B4/B5 Wi-Fi: 802.11 b/g/n I/O: 2× configurable DI/DO

Model	Region	Description
ODU302-EUC4	Europe/APAC	LTE-FDD: B1/B3/B5/B7/B8/B20/B28 LTE-TDD: B38/B40/B41 WCDMA: B1/B5/B8 GSM/EDGE: B3/B8 Wi-Fi: 802.11 b/g/n I/O: 2× configurable DI/DO

10.3 Package Contents

- ODU302 outdoor unit × 1
- IEEE 802.3af PoE power adapter × 1
- 1 m RJ45 cable × 1
- Mounting kit (wall/pole/top-mount) × 1
- Quick Start Guide × 1

11. Reliability Standards & Certifications

11.1 Protection Rating

- IP Rating: IP65 (dust-tight, protected against high-pressure water jets)

11.2 Mechanical Reliability

Test Item	Standard
Shock	IEC 60068-2-27
Vibration	IEC 60068-2-6
Free Fall	IEC 60068-2-32

11.3 EMC Immunity

Test Item	Standard
ESD	EN 61000-4-2, Level 2
Radiated RF	EN 61000-4-3, Level 2
EFT/Burst	EN 61000-4-4, Level 2
Surge	EN 61000-4-5, Level 2
Conducted RF	EN 61000-4-6, Level 2
Oscillatory Wave	EN 61000-4-12, Level 2
Power Frequency Magnetic	EN 61000-4-8, 400 A/m

11.4 Certifications & Compliance

Certification	Status
FCC	Completed
IC	Completed
PTCRB	Completed
AT&T	Completed
Verizon	Completed
T-Mobile	Completed

Note: Certification status is current as of publication. Check with InHand sales for the latest updates.

12. Contact Us

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